COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / DATA STRUCTURES AND ALGORITHMS	**SECTION:**	CPE21S4
DATE PERFORMED:	07 / 15 / 26	**DATE SUBMITTED:**	07 / 15 / 26
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ACTIVITY NO. 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

ILO A:

```cpp
#include <iostream>
using namespace std;

void sum(int a, int b);
int greaterNum(int A, int B);
bool logicalOperation(bool a, bool b);

int main() {
    int choice;

    cout << "MENU" << endl;
    cout << "1. Sum of Two Numbers" << endl;
    cout << "2. Greater Number" << endl;
    cout << "3. Logical Operations" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
    case 1: {
        int num1, num2;
        cout << "\nEnter first number: ";
        cin >> num1;
        cout << "Enter second number: ";
        cin >> num2;
        sum(num1, num2);
        break;
    }

    case 2: {
        int A, B;
        cout << "\nEnter value of A: ";
        cin >> A;
        cout << "Enter value of B: ";
        cin >> B;
        cout << "Greater value: " << greaterNum(A, B) << endl;
        break;
    }
```

```
    case 3: {
        bool x, y;
        cout << "\nEnter first value (1 or 0): ";
        cin >> x;
        cout << "Enter second value (1 or 0): ";
        cin >> y;
        logicalOperation(x, y);
        break;
    }

    default:
        cout << "Invalid choice!" << endl;
    }

    return 0;
}


void sum(int a, int b) {
    cout << "Sum: " << a + b << endl;
}

int greaterNum(int A, int B) {
    if (A > B)
        return A;
    else
        return B;
}

bool logicalOperation(bool a, bool b) {
    cout << "AND: " << (a && b) << endl;
    cout << "OR: " << (a || b) << endl;
    cout << "NOT A: " << (!a) << endl;
    cout << "NOT B: " << (!b) << endl;

    return true;
}
```

OUTPUT:

```
MENU
1. Sum of Two Numbers
2. Greater Number
3. Logical Operations
Enter your choice: 1

Enter first number: 10
Enter second number: 15
Sum: 25

--------------------------------
Process exited after 10.86 seconds with return value 0
Press any key to continue . . .
```

```
MENU
1. Sum of Two Numbers
2. Greater Number
3. Logical Operations
Enter your choice: 2

Enter value of A: 5
Enter value of B: 15
Greater value: 15

--------------------------------
Process exited after 7.07 seconds with return value 0
Press any key to continue . . .
```

```
MENU
1. Sum of Two Numbers
2. Greater Number
3. Logical Operations
Enter your choice: 3

Enter first value (1 or 0): 1
Enter second value (1 or 0): 1
AND: 1
OR: 1
NOT A: 0
NOT B: 0

--------------------------------
Process exited after 15.63 seconds with return value 0
Press any key to continue . . .
```

ILO B:

```cpp
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle.";
    }
    else {
        cout << "The shape is NOT a valid triangle.";
    }

    return 0;
}
```

OUTPUT:

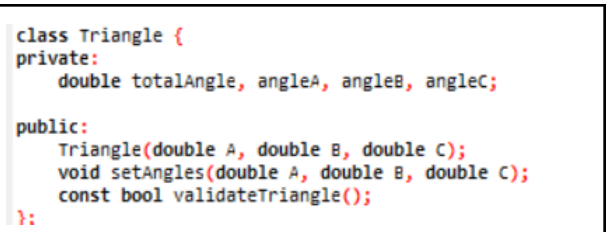
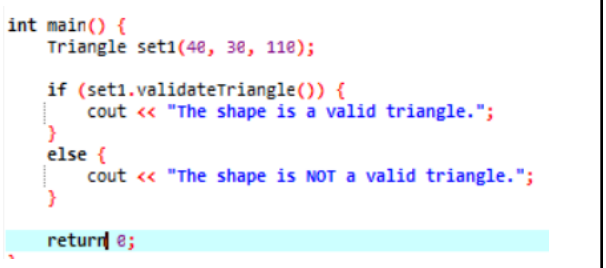
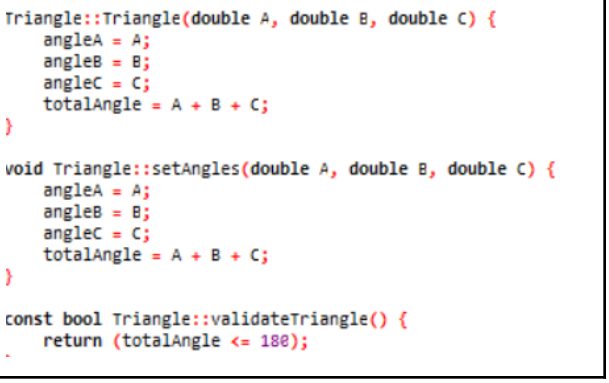
(A)

```
The shape is a valid triangle.
--------------------------------
Process exited after 0.1299 seconds with return value 0
Press any key to continue . . .
```

(B)

```
The shape is NOT a valid triangle.
--------------------------------
Process exited after 0.1363 seconds with return value 0
Press any key to continue . . .
```

Sections	Answer
Header File Declaration Section	**#include <iostream>** **using namespace std;**
Global Declaration Section	**None**

Class Declaration and Method Definition Section	
Main Function	
Method Definition	

```
class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
```

```
int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle.";
    }
    else {
        cout << "The shape is NOT a valid triangle.";
    }

    return 0;
```

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
```

ILO C: (1, 2, 3)

```
#include <iostream>
#include <cmath>
using namespace std;

void swapNumbers(int &, int &);
double kelvinToFahrenheit(double);
double distance(double, double, double, double);

int main()
{
    int choice;

    cout << "MENU" << endl;
    cout << "1. Swap Two Numbers" << endl;
    cout << "2. Kelvin to Fahrenheit" << endl;
    cout << "3. Distance Between Two Points" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    if (choice == 1)
    {
        int num1, num2;

        cout << "Enter first number: ";
        cin >> num1;

        cout << "Enter second number: ";
        cin >> num2;

        cout << "Before Swapping:" << endl;
        cout << "First Number = " << num1 << endl;
        cout << "Second Number = " << num2 << endl;

        swapNumbers(num1, num2);
```

```
        cout << "After Swapping:" << endl;
        cout << "First Number = " << num1 << endl;
        cout << "Second Number = " << num2 << endl;
    }
    else if (choice == 2)
    {
        double kelvin, fahrenheit;

        cout << "Enter temperature in Kelvin: ";
        cin >> kelvin;

        fahrenheit = kelvinToFahrenheit(kelvin);

        cout << "Temperature in Fahrenheit = " << fahrenheit << endl;
    }
    else if (choice == 3)
    {
        double x1, y1, x2, y2, result;

        cout << "Enter x1: ";
        cin >> x1;

        cout << "Enter y1: ";
        cin >> y1;

        cout << "Enter x2: ";
        cin >> x2;

        cout << "Enter y2: ";
        cin >> y2;

        result = distance(x1, y1, x2, y2);

        cout << "Distance = " << result << endl;
    }
    {
        cout << "Invalid choice!" << endl;
    }

    return 0;
}


void swapNumbers(int &a, int &b)
{
    int temp;

    temp = a;
    a = b;
    b = temp;
}

double kelvinToFahrenheit(double k)
{
    return (k - 273.15) * 9 / 5 + 32;
}

double distance(double x1, double y1, double x2, double y2)
{
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}
```

OUTPUTS:

1.

```
MENU
1. Swap Two Numbers
2. Kelvin to Fahrenheit
3. Distance Between Two Points
Enter your choice: 1
Enter first number: 124
Enter second number: 1645
Before Swapping:
First Number = 124
Second Number = 1645
After Swapping:
First Number = 1645
Second Number = 124

--------------------------------
Process exited after 9.87 seconds with return value 0
Press any key to continue . . .
```

2.

```
MENU
1. Swap Two Numbers
2. Kelvin to Fahrenheit
3. Distance Between Two Points
Enter your choice: 2
Enter temperature in Kelvin: 100
Temperature in Fahrenheit = -279.67

--------------------------------
Process exited after 3.212 seconds with return value 0
Press any key to continue . . .
```

3.

```
MENU
1. Swap Two Numbers
2. Kelvin to Fahrenheit
3. Distance Between Two Points
Enter your choice: 3
Enter x1: 8
Enter y1: 3
Enter x2: 4
Enter y2: -5
Distance = 8.94427

--------------------------------
Process exited after 8.65 seconds with return value 0
Press any key to continue . . .
```

ILO C: (4)

```cpp
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();

    double area();
    double perimeter();
    void triangleType();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

double Triangle::area() {
    return (5 * 4) / 2;
}

double Triangle::perimeter() {
    return 5 + 6 + 7;
}

void Triangle::triangleType() {
    if (angleA < 90 && angleB < 90 && angleC < 90)
        cout << "Triangle is Acute-angled.";
    else if (angleA > 90 || angleB > 90 || angleC > 90)
        cout << "Triangle is Obtuse-angled.";
    else
        cout << "Triangle is Others.";
}

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle." << endl;
        cout << "Area: " << set1.area() << endl;
        cout << "Perimeter: " << set1.perimeter() << endl;
        set1.triangleType();
    }
    else {
        cout << "The shape is NOT a valid triangle.";
    }

    return 0;
}
```

OUTPUT:

```
int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle." << endl;
        cout << "Area: " << set1.area() << endl;
        cout << "Perimeter: " << set1.perimeter() << endl;
        set1.triangleType();
```

```
C:\Users\Rhianne\Downloads\
The shape is a valid triangle.
Area: 10
Perimeter: 18
Triangle is Obtuse-angled.
```

```
Triangle set1(70, 50, 60);

if (set1.validateTriangle()) {
    cout << "The shape is a valid triangle." << endl;
    cout << "Area: " << set1.area() << endl;
```

```
The shape is a valid triangle.
Area: 10
Perimeter: 18
Triangle is Acute-angled.
```

```
main() {
Triangle set1(30, 60, 90);
```

```
The shape is a valid triangle.
Area: 10
Perimeter: 18
Triangle is Others.
```

NOTE:
I had difficulty modifying the given Triangle class because the original code only used angles, while the area and perimeter required additional values. I also found it challenging to add new functions without changing too much of the original code. Despite these challenges, I was able to complete the activity after reviewing and practicing the concepts.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. CONCLUSION

I think I did well in this activity because I was able to complete the required programs and understand how functions and classes are used in C++. However, I had some difficulty modifying the given Triangle class, especially when adding the functions for the area and perimeter because the original code only used angles. I also found it challenging to determine what changes were needed while keeping the original code simple. Overall, this activity helped improve my problem-solving and programming skill.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: ChatGPT and QuillBot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Grammar correction and Explanation of Concepts

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: ChaGPT was used to search for and understand concepts, while QuillBot was used to check and correct grammar.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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